

A generalization of compact spaces;

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English translation

The importance acquired in recent work in topology by the notion of an *open covering* of a topological space is well known. The predominant position occupied in topology by *compact*¹ spaces is due to the fundamental property of their open coverings expressed by the Borel-Lebesgue axiom: from every open covering one can extract a *finite* covering, that is, one consisting of finitely many sets. In this paper we begin the study of a broader class of topological spaces, characterized by a less restrictive property of their open coverings. In particular, we show that these spaces, which we propose to call *paracompact*, are *normal* spaces; conversely, a normal space is not always paracompact, but the most usual spaces are, for example metric spaces with a countable base². Moreover, these spaces have properties that further resemble those of compact spaces with regard to the *uniform structures* compatible with their topology.

¹In this paper we use the terminology and notation of N. Bourbaki's *Éléments de Mathématique* (*Actual. Scient. et Ind.*, nos. 846, 858, 916 and 934). In particular, the spaces we call *compact* are those called *bicompact* in the terminology of Alexandroff and Urysohn: the development of general topology and its applications to analysis shows increasingly that this is the fundamental notion, rather than that of a "compact" space in the sense of M. Fréchet; indeed, spaces which are "compact" in M. Fréchet's sense but not "bicompact" are for the most part pathological. This is why an increasing number of mathematicians, following N. Bourbaki, reserve the name "compact" space for those which are truly useful in mathematics, namely the "bicompact" spaces of Alexandroff and Urysohn.

²As is well known, these spaces were called "separable" by M. Fréchet. This terminology is rather unfortunate, since, on the one hand, it suggests no image corresponding to the notion it denotes and, on the other, it risks causing regrettable confusion with the "separation axioms" of topological spaces, which concern an entirely different order of questions. It would therefore be desirable for the use of this term to be abandoned altogether.

1. By a *covering* of a topological space E , we shall always mean below an *open* covering, that is, a family $\mathcal{R}$ of open sets A_α of E (where the index α ranges over a set of arbitrary cardinality) such that E is the *union* of the sets A_α . We shall say that a covering $\mathcal{R}$ is *point-finite* if each point of E belongs to only *finitely* many sets of the covering $\mathcal{R}$; we shall say that $\mathcal{R}$ is *locally finite*³ if, for every point $x \in E$, there exists a neighborhood V of x meeting only *finitely* many sets of $\mathcal{R}$. A locally finite covering is clearly point-finite, but the converse is false⁴.

Further, given two coverings $\mathcal{R}, \mathcal{R}'$ of E , we shall say that $\mathcal{R}$ is *subordinate* to $\mathcal{R}'$ if every set of the covering $\mathcal{R}$ is *contained* in a set of the covering $\mathcal{R}'$ (thus we do not assume that $\mathcal{R}$ is *extracted* from $\mathcal{R}'$, that is, that every set of $\mathcal{R}$ is also a set of $\mathcal{R}'$).

With this understood, we shall call a *paracompact space* any *Hausdorff* topological space E satisfying the following axiom:

For every covering $\mathcal{R}$ of E , there exists a locally finite covering $\mathcal{R}'$ of E subordinate to $\mathcal{R}$.

2. THEOREM 1. — *Every paracompact space is normal.*

We proceed in two steps, showing successively that a paracompact space E : 1° is regular; 2° is normal.

1° E is *regular*. Indeed, let F be a closed subset of E , and a a point not belonging to F . For every $x \in F$, there exist by hypothesis (E being

³This definition is slightly less restrictive than that given by A. Weil (for locally compact spaces only) in his work *Sur les espaces à structure uniforme*, p. 34 (*Actual. Scient. et Ind.*, no. 551).

⁴Here is an example of a point-finite covering which is not locally finite: let E be the set consisting of the integers > 0 and the first transfinite number ω of the second class, and F the set consisting of the transfinite numbers of the first and second classes and the first transfinite number Ω of the third class. Give E (resp. F) the topology in which, for every $x \neq \omega$ (resp. $x \neq \Omega$), the set $\{x\}$ is a neighborhood of x , and a neighborhood base at ω (resp. Ω) consists of the closed intervals with initial point x and final point ω (resp. Ω), where x ranges over all numbers $< \omega$ (resp. $< \Omega$). Let G be the product space of E and F , and H the subspace of G complementary to (ω, Ω) . Let $\mathcal{R}$ be a covering of H obtained by taking, for each point (x, Ω) of H , the product of $\{x\}$ with a neighborhood of Ω ; for each point (ω, y) , the product of a neighborhood of ω with $\{y\}$; and for the points (x, y) outside the union of these neighborhoods, the set $\{(x, y)\}$. It is immediate that every point of H belongs to at most two sets of $\mathcal{R}$; but there exists an ordinal α such that, for every $y > \alpha$, every neighborhood of (ω, y) meets infinitely many sets of $\mathcal{R}$ which are neighborhoods of points (x, Ω) .

Hausdorff) an open neighborhood V_x of x and an open neighborhood W_x of a which are disjoint. Consider the covering $\mathcal{R}$ of E formed by the V_x , where x ranges over F , and the complement F' of F . There exists a locally finite covering $\mathcal{R}'$ subordinate to $\mathcal{R}$; the union U of those sets of $\mathcal{R}'$ which meet F is an open set containing F . We shall show that there exists a neighborhood V of a disjoint from U , which will prove the assertion. By hypothesis, there exists a neighborhood W of a meeting only finitely many sets $A_1, A_2, \dots, A_n$ of $\mathcal{R}'$; each of the A_i which meets F is, by definition, contained in some V_{x_i} corresponding to a point $x_i \in F$. If we take V to be the intersection of W and the W_{x_i} corresponding to these points, then V is a neighborhood of a meeting none of those A_i which meet F ; *a fortiori*, V does not meet U .

2° E is *normal*. Indeed, let F and G be two disjoint closed sets in E . For every $x \in F$, there exist an open neighborhood V_x of x and an open neighborhood W_x of G which are disjoint (E being regular); consider the covering $\mathcal{R}$ of E formed by the V_x , where x ranges over F , and the complement F' of F . Let $\mathcal{R}'$ be a locally finite covering subordinate to $\mathcal{R}$. The union U of those sets of $\mathcal{R}'$ which meet F is an open set containing F ; we shall show that there exists a neighborhood V of G disjoint from U , which will complete the proof. For every $y \in G$, there exists by hypothesis a neighborhood T_y of y meeting only finitely many sets $A_1, A_2, \dots, A_n$ of $\mathcal{R}'$; each of those A_i which meets F is by definition contained in some V_{x_i} corresponding to a point $x_i \in F$. Let S_y be the intersection of T_y and the W_{x_i} corresponding to these points; S_y is a neighborhood of y disjoint from U . If V is the union of the S_y as y ranges over G , then V is a neighborhood of G disjoint from U .

3. Theorem 1 immediately raises the converse question: is every normal space paracompact? We shall see from an example that this is not so.

Let E be the set of ordinal numbers of the first and second classes, equipped with the topology in which a neighborhood base at a point x consists of the intervals $[y, x]$ ending at that point and such that $y < x$. It is known⁵ that this space is *locally compact and completely normal*. Consider a covering $\mathcal{R}$ of E consisting of *nonempty bounded open intervals*; we shall see that there is no *point-finite* covering $\mathcal{R}'$ subordinate to $\mathcal{R}$. Indeed, otherwise, for

⁵For these well-known properties, see, for example, J. DIEUDONNÉ, *Un exemple d'espace normal non susceptible d'une structure uniforme d'espace complet* (C. R. Acad. Sc., vol. **209**, 1938, pp. 145–147).

every $x \in E$, the intersection of the sets of $\mathcal{R}'$ containing x would be a neighborhood of x , and hence would contain an interval $[y_x, x]$ with $y_x < x$; on the other hand, no point z could belong to infinitely many intervals $[y_x, x]$, so there would exist a z' such that $x \geq z'$ implies $y_x \geq z$. But this conclusion is absurd⁽⁵⁾.

4. Every compact space is clearly paracompact; to obtain other examples of paracompact spaces, we begin by establishing the following theorem:

THEOREM 2. — *a. In a paracompact space E , every closed subspace is paracompact.*

b. If every open subspace of a paracompact space E is paracompact, then every subspace of E is paracompact.

a. Let F be a closed subspace of a paracompact space E , and $\mathcal{R}$ a covering of F . The sets A_α of $\mathcal{R}$ are open in F , each being the trace on F of a set U_α open in E ; let $\mathcal{R}_1$ be the open covering of E formed by the U_α and the complement F' of F . By hypothesis, there exists a locally finite covering $\mathcal{R}'_1$ of E subordinate to $\mathcal{R}_1$. If a set of this covering meets F , it is necessarily contained in some U_α , so its trace on F is contained in some A_α ; hence the traces on F of the sets of $\mathcal{R}'_1$ form a covering $\mathcal{R}'$ of F which is clearly locally finite and subordinate to $\mathcal{R}$.

b. Let G be a subspace of a paracompact space E , and $(A_\alpha) = \mathcal{R}$ a covering of G ; each A_α is the trace on G of a set U_α open in E . Let U be the union of the U_α . Then U is an open subspace of E , and the U_α form a covering $\mathcal{R}_1$ of U . If U is paracompact, there exists a locally finite covering $\mathcal{R}'_1$ of U subordinate to $\mathcal{R}_1$; the traces on G of the sets of $\mathcal{R}'_1$ clearly form a locally finite covering of G subordinate to $\mathcal{R}$.

5. Recall that a *locally compact* space E is called σ -compact if it is a countable union of compact sets⁶.

THEOREM 3. — *Every locally compact σ -compact space E is paracompact.*

⁶Cf. A. WEIL, *loc. cit.*, p. 28; the following proof is related to A. Weil's arguments there and on p. 35 of the same work.

First let us show that E is the union of a sequence of relatively compact open sets U_n such that $\overline{U}_n \subset U_{n+1}$. By hypothesis, E is the union of countably many compact sets K_n , and we may always assume that these sets form an increasing sequence (replacing K_n , if necessary, by the union of the K_m with indices $\leq n$); we shall define the U_n inductively so that $K_n \subset U_n$. To this end, note that every point of E has a relatively compact open neighborhood. Covering K_1 by such neighborhoods, we can extract a finite covering of K_1 ; if U_1 is the union of the sets of this covering, then U_1 is relatively compact. Now suppose that U_m has been defined with $K_m \subset U_m$ for $m \leq n$; similarly, the compact set $\overline{U}_n \cup K_{n+1}$ can be covered by finitely many relatively compact open neighborhoods. Their union U_{n+1} is a relatively compact open set such that $K_{n+1} \subset U_{n+1}$ and $\overline{U}_n \subset U_{n+1}$.

Now let $\mathcal{R} = (A_\alpha)$ be any covering of E . Denote by V_n the compact set consisting of the points of $\overline{U}_n$ not belonging to U_{n-1} . For every point $x \in V_n$, there exists a neighborhood W_x of x contained in one of the A_α , also contained in U_{n+1} (since $\overline{U}_n \subset U_{n+1}$), and disjoint from $\overline{U}_{n-2}$ (since $\overline{U}_{n-2} \subset U_{n-1}$); we can cover V_n by *finitely* many of the W_x . Proceed similarly for each index n , and let $\mathcal{R}'$ be the resulting covering of E ; clearly $\mathcal{R}'$ is subordinate to $\mathcal{R}$. We shall see that it is *locally finite*. Indeed, let y be any point of E , and n the smallest integer such that $y \in U_n$. Since y is not in U_{n-1} , there exists a neighborhood T of y contained in U_n and disjoint from $\overline{U}_{n-2}$; hence T can meet only those sets of $\mathcal{R}'$ arising from the coverings of V_{n-2} , V_{n-1} , V_n or V_{n+1} , and there are only finitely many such sets.

COROLLARY. — *Every locally compact space E with a countable base is paracompact*⁷.

Indeed, let (U_n) be a base of E . For every point x of E , there exists a compact neighborhood of x , and hence some U_n containing x and contained in this neighborhood, and thus relatively compact. Keeping only those U_n which are relatively compact, we therefore still have a base of E . But then E is the union of the compact sets $\overline{U}_n$, and is therefore σ -compact.

⁷More generally, every *locally compact metrizable* space is paracompact, since such a space is a topological sum (BOURBAKI, *Top. gén.*, chap. I, § 8) of locally compact spaces with countable bases, by a theorem of P. Alexandroff (*Math. Ann.*, vol. **92**, 1924, pp. 299–301).

6. The preceding corollary can be generalized by combining it with Theorem 2:

THEOREM 4. — *Every metrizable space E with a countable base is paracompact.*

Indeed, a classical theorem of Urysohn shows that such a space may be regarded as embedded in a compact metrizable space K . Every *open* subspace of K is locally compact and has a countable base; Theorem 2 and the corollary to Theorem 3 therefore show that every subspace of K is paracompact.

7. There are paracompact spaces which are not compact and do not have countable bases⁽⁷⁾. This follows, for example, from the next theorem:

THEOREM 5. — *The product of a compact space and a paracompact space is paracompact.*

Let E be a compact space, F a paracompact space, and $\mathcal{R}$ any covering of the product space $E \times F$. For every point $(x, y) \in E \times F$, there exist a neighborhood $V_{x,y}$ of x in E and a neighborhood $W_{x,y}$ of y in F such that the neighborhood $V_{x,y} \times W_{x,y}$ of (x, y) is contained in a set of $\mathcal{R}$. For fixed y in F , the sets $V_{x,y} \times W_{x,y}$ (where x ranges over E) form a covering of the compact set $E \times \{y\}$. Thus there are finitely many points $x_i \in E$ such that the $V_{x_i,y} \times W_{x_i,y}$ still cover $E \times \{y\}$. If W_y is the intersection of the $W_{x_i,y}$, the sets $V_{x_i,y} \times W_y$ therefore still cover $E \times \{y\}$, and W_y is a neighborhood of y in F . Now let $\mathcal{R}_1$ be the covering of F formed by the W_y ; there exists a locally finite covering $\mathcal{R}'_1$ of F subordinate to $\mathcal{R}_1$. Each set A_α of $\mathcal{R}'_1$ is contained in some W_y ; choose for each α a y_α with this property, and consider the sets $V_{x_i,y_\alpha} \times A_\alpha$. These form a covering $\mathcal{R}'$ of $E \times F$ which is clearly subordinate to $\mathcal{R}$. Let us show that $\mathcal{R}'$ is locally finite. If (a, b) is any point of $E \times F$, there exists a neighborhood U of b meeting only finitely many sets of the covering $\mathcal{R}'_1$; by the definition of $\mathcal{R}'$, the neighborhood $E \times U$ of (a, b) meets only finitely many sets of $\mathcal{R}'$.

8. We shall now study certain uniform structures compatible with the topology of a paracompact space. First, we prove a property of *point-finite* coverings of a normal space which generalizes a well-known result concerning *finite* coverings of such spaces:

THEOREM 6. — *Let (A_α) be a point-finite covering of a normal space E . There exists a covering (B_α) of E with the same index set, such that $\overline{B_\alpha} \subset A_\alpha$ for every index α .*

Consider all coverings $\mathcal{R} = (X_\alpha)$ of E with the same index set I as (A_α) and with the following property: there exists a subset $H_{\mathcal{R}}$ of I such that, for $\alpha \in H_{\mathcal{R}}$, $\overline{X_\alpha} \subset A_\alpha$, and, for $\alpha \notin H_{\mathcal{R}}$, $X_\alpha = A_\alpha \neq \overline{A_\alpha}$. Let Φ be the set of all these coverings. It is nonempty, since the given covering $\mathcal{R}_0 = (A_\alpha)$ belongs to Φ , taking $H_{\mathcal{R}_0}$ to be the set of indices α such that A_α is both open and closed. We shall order the set Φ : if $\mathcal{R} = (X_\alpha)$ and $\mathcal{R}' = (Y_\alpha)$, put $\mathcal{R} \leq \mathcal{R}'$ if $H_{\mathcal{R}} \subset H_{\mathcal{R}'}$ and if $Y_\alpha = X_\alpha$ for every $\alpha \in H_{\mathcal{R}}$; this is clearly an order relation. Let us show that Φ , with this order, is *inductive*, that is, every chain has an upper bound. Indeed, let Ψ be a *totally ordered* subset of Φ , and let K be the subset of I which is the union of the $H_{\mathcal{R}}$ as $\mathcal{R}$ ranges over Ψ . For $\alpha \in K$, denote by Z_α the set with index α in any one of the coverings $\mathcal{R}$ such that $\alpha \in H_{\mathcal{R}}$ (this set is the same for all such coverings, by hypothesis); if instead $\alpha \notin K$, take $Z_\alpha = A_\alpha$. Let us show that $\mathcal{C} = (Z_\alpha)$ is a covering of E . There are only finitely many indices α such that $x \in A_\alpha$; if one of these indices does not belong to K , then $x \in Z_\alpha$. If, on the contrary, all belong to K , they necessarily belong to one and the same $H_{\mathcal{R}}$ (since Ψ is totally ordered). Since $\mathcal{R} = (X_\alpha)$ is a covering of E , x belongs to at least one of the X_α corresponding to these indices, and since the corresponding Z_α are respectively equal to the X_α , x belongs to at least one Z_α . It is then clear that $\mathcal{C} = K$, and that $\mathcal{C}$ is the supremum of Ψ in Φ , proving that Φ is inductive. By Zorn's theorem, Φ has a *maximal* element $\mathcal{M} = (M_\alpha)$; we shall see that $H_{\mathcal{M}} = I$, which will prove the theorem. Otherwise, let α be an index not belonging to $H_{\mathcal{M}}$; thus $M_\alpha = A_\alpha$. Let U be the open set which is the union of the M_β with indices $\beta \neq \alpha$; the complement N_α of M_α is a closed set contained in U . Since E is normal, there exists an open set V such that $N_\alpha \subset V$ and $\overline{V} \subset U$. If M'_α denotes the complement of $\overline{V}$, then $\overline{M'_\alpha} \subset A_\alpha$, and M'_α , together with the M_β of indices $\beta \neq \alpha$, forms a covering $\mathcal{R}$ of E belonging to Φ and $> \mathcal{M}$, contrary to the definition of $\mathcal{M}$.

9. If $\mathcal{R} = (A_\alpha)$ is a covering of a topological space E , the union $V_{\mathcal{R}}$ of the sets $A_\alpha \times A_\alpha$ is a neighborhood of the diagonal Δ in the product space $E \times E$.

THEOREM 7. — *For the sets $V_{\mathcal{R}}$ corresponding to all locally finite coverings $\mathcal{R}$ of a Hausdorff space E to form a base of entourages of a uniform structure compatible with the topology of E , it is necessary and sufficient that E be normal.*

1° The condition is *necessary*. Indeed, let F and G be two disjoint closed sets in E , and A and B the complements of F and G , respectively; A and B form a finite covering $\mathcal{R} = (A, B)$ of E . Suppose there exists a symmetric neighborhood W of the diagonal Δ of $E \times E$ such that $\overset{2}{W} \subset V_{\mathcal{R}}$. By the definition of $V_{\mathcal{R}}$, for every $x \in F$ we have

$$\overset{2}{W}(x) \subset V_{\mathcal{R}}(x) = B;$$

similarly, for every $y \in G$, we have

$$\overset{2}{W}(y) \subset V_{\mathcal{R}}(y) = A;$$

but since W is symmetric, we conclude that $W(x) \cap W(y)$ is empty, and hence so is $W(F) \cap W(G)$. Since $W(F)$ is a neighborhood of F and $W(G)$ a neighborhood of G , we see that E is normal.

2° The condition is *sufficient*. First note that, as $\mathcal{R}$ ranges over the locally finite coverings of E , the $V_{\mathcal{R}}$ form a filter base on $E \times E$. For if $\mathcal{R} = (A_{\alpha})$ and $\mathcal{R}' = (B_{\beta})$ are two such coverings, the sets $A_{\alpha} \cap B_{\beta}$ form a third covering $\mathcal{R}''$ of E which is clearly locally finite, and $V_{\mathcal{R}''} \subset V_{\mathcal{R}} \cap V_{\mathcal{R}'}$. To see that this filter base is a base of entourages of a uniform structure $\mathfrak{U}$, it therefore suffices⁸ to prove that, for every locally finite covering $\mathcal{R} = (A_{\alpha})$, there exists a locally finite covering $\mathcal{R}' = (B_{\beta})$ such that $\overset{2}{V}_{\mathcal{R}'} \subset V_{\mathcal{R}}$.

Apply Theorem 6 twice in succession to the covering $\mathcal{R}$, forming two new locally finite coverings (A'_{α}) and (A''_{α}) such that, for every index α , $\overline{A'_{\alpha}} \subset A_{\alpha}$ and $\overline{A''_{\alpha}} \subset A'_{\alpha}$. For each α , denote by B'_{α} the complement of $\overline{A'_{\alpha}}$ and by B''_{α} the complement of $\overline{A''_{\alpha}}$; then consider the family $\mathcal{F}$ consisting of the open sets A'_{α} , $A_{\alpha} \cap B''_{\alpha}$ and B'_{α} (for all indices α). Let x be any point of E , and W_x the intersection of all sets of $\mathcal{F}$ containing x ; first let us show that W_x is *open*. Since $W_x \subset W_y$ for every $y \in W_x$, it suffices to show that W_x is a neighborhood of x . By hypothesis, there exists a neighborhood U of x meeting only finitely many of the A_{α} ; for those α such that A_{α} does not meet U , U is therefore contained in B'_{α} . Thus the intersection of U and W_x is the intersection of U and finitely many open sets, so W_x is indeed a neighborhood of x . Now let $\mathcal{R}'$ be the covering formed by the distinct sets W_x ; an argument similar to the preceding one immediately shows that $\mathcal{R}'$ is locally finite (since for every point $x \in E$ there exists a neighborhood of x contained in all but finitely many of the B'_{α}). If,

⁸ Cf. BOURBAKI, *Top. gén.*, chap. II, § 1.

for simplicity, we put $H = V_{\mathcal{R}}$, $K = V_{\mathcal{R}'}$, we shall show that $\overset{2}{K}(z) \subset H(z)$ for every $z \in E$, and hence $\overset{2}{K} \subset H$, which will establish the assertion.

By hypothesis, there exists an index α such that $z \in A''_{\alpha}$. If $u \in K(z)$, then by definition there exists an x such that $z \in W_x$ and $u \in W_x$. Necessarily $x \in A'_{\alpha}$, for otherwise, by the definition of W_x , this set would be contained in B''_{α} and hence could not contain z . It follows that $u \in A'_{\alpha}$ as well. Now if $v \in K(u)$, there similarly exists a y such that $u \in W_y$ and $v \in W_y$. Necessarily $y \in A_{\alpha}$, for otherwise W_y would be contained in B'_{α} and hence would not contain u . We conclude that $v \in A_{\alpha}$, or in other words that $\overset{2}{K}(z) \subset A_{\alpha} \subset H(z)$.

It remains to prove that the uniform structure $\mathfrak{U}$ defined by the entourages $V_{\mathcal{R}}$ is compatible with the topology of E . First note that $V_{\mathcal{R}}(x)$ is always a neighborhood of x ; conversely, if U is any open neighborhood of x , there exists a closed neighborhood F of x contained in U . If W denotes the complement of F , and $\mathcal{R}$ the finite covering (U, W) of E , then $V_{\mathcal{R}}(x) = U$, completing the proof⁹.

10. Given a completely regular space E , it is known¹⁰ that among all uniform structures compatible with the topology of E , there

⁹This argument also shows that for the sets $V_{\mathcal{R}}$ corresponding to *finite* coverings $\mathcal{R}$ of E to form a base of entourages of a uniform structure compatible with the topology of E , it is again necessary and sufficient that E be *normal*. We communicated this result and its proof to N. Bourbaki in 1937 without publishing it; the theorem was obtained independently by K. Morita [*On uniform Spaces and the dimension of compact spaces* (*Proc. Phys. Math. Soc. of Japan*, 1940, pp. 969–977)]. Here is a third proof whose principle is close to that of Morita's. Since E is normal, and therefore completely regular, the coarsest uniform structure $\mathfrak{U}$ making the bounded continuous numerical functions on E uniformly continuous is compatible with the topology of E and makes E precompact; hence its completion $\beta(E) = F$ for this uniform structure is a *compact* space. We shall show that the $V_{\mathcal{R}}$ define the structure $\mathfrak{U}$. First, an entourage U of the structure $\mathfrak{U}$ is the trace on $E \times E$ of an entourage U_0 of the unique uniform structure of the compact space F ; U_0 contains a union of sets $A_i \times A_i$, where (A_i) is a finite covering of F . The traces of the A_i on E form a finite covering $\mathcal{R}$ of E , and clearly $V_{\mathcal{R}} \subset U$.

Conversely, we shall see that every $V_{\mathcal{R}}$ contains an entourage of the structure $\mathfrak{U}$. Let $\mathcal{R} = (B_i)$, and let $\mathcal{R}' = (C_i)$ be a covering of E such that $\overline{C_i} \subset B_i$ for every index i . There exists, by Urysohn's theorem, a continuous function f_i on E , equal to 0 on $\overline{C_i}$, to 1 at points of the complement of B_i , and taking values between 0 and 1 at the other points of E . Consider the entourage U of the structure $\mathfrak{U}$ consisting of pairs (x, y) such that $|f_i(x) - f_i(y)| < 1$ for every index i . For any x in E , there exists an index i such that $x \in C_i$, and hence $U(x) \subset B_i \subset V_{\mathcal{R}}(x)$ (since the latter set is the union of all B_j to which x belongs), proving that $U \subset V_{\mathcal{R}}$.

¹⁰A. WEIL, *loc. cit.*, p. 16.

is one, $\mathfrak{U}_0$, finer than all the others (the *universal structure* of E). It is natural to seek a simple description of the filter of entourages of this structure. Since every entourage of a uniform structure compatible with the topology of E is a neighborhood of Δ , the filter of entourages of such a structure is always coarser than the filter $\mathfrak{V}$ of neighborhoods of Δ in $E \times E$. We are thus led to ask whether $\mathfrak{V}$ might itself be the filter of entourages of a uniform structure compatible with the topology of E , in which case that structure would necessarily be the universal structure $\mathfrak{U}_0$. But the first part of the proof of Theorem 7 shows that $\mathfrak{V}$ can be the filter of entourages of a uniform structure only if E is *normal*. We do not know whether, conversely, this condition is sufficient; but we have the following result:

THEOREM 8. — *If E is a paracompact space, the filter of neighborhoods of the diagonal Δ in $E \times E$ is the filter of entourages of the universal structure of E .*

Let U be any neighborhood of Δ ; it suffices to prove that there exists a neighborhood W of Δ such that $\overset{2}{W} \subset U$. For every $x \in E$, let A_x be an open neighborhood of x such that $A_x \times A_x \subset U$; the A_x form a covering $\mathcal{R}$ of E . Let $\mathcal{R}'$ be a locally finite covering subordinate to $\mathcal{R}$; then $V_{\mathcal{R}'} \subset U$. By Theorem 7, there exists a locally finite covering $\mathcal{R}''$ of E such that $\overset{2}{V}_{\mathcal{R}''} \subset V_{\mathcal{R}'}$; putting $W = V_{\mathcal{R}''}$, we obtain a W with the required properties. This proof also shows that the $V_{\mathcal{R}}$ form a base for the filter of entourages of $\mathfrak{U}_0$ as $\mathcal{R}$ ranges over all locally finite coverings of E .

11. The following theorem generalizes to numerical functions defined on a paracompact space a classical property of numerical functions defined on a compact space.

THEOREM 9. — *Let E be a paracompact space, g a lower semicontinuous numerical function on E , and h an upper semicontinuous numerical function on E , such that $h(x) < g(x)$ at every point $x \in E$. Then there exists a numerical function f , continuous on E , such that $h(x) < f(x) < g(x)$ at every point of E .*

By hypothesis, for every $x \in E$, there exist an open neighborhood U_x of x and a number a_x such that $h(y) < a_x < g(y)$ at every point $y \in U_x$. Let (A_α) be a locally finite covering subordinate to the covering formed by the U_x , and then (Theorem 6) let (B_α) be a covering of E such that $\overline{B}_\alpha \subset A_\alpha$ for every index α . For each index α , there exists a numerical function f_α , continuous on E , equal to $-\infty$ on the complement of A_α , to a_{x_α} on $\overline{B}_\alpha$,

and lying between $-\infty$ and a_{x_α} at the other points of E ; here x_α denotes a point such that A_α is contained in U_{x_α} . Thus $f_\alpha(x) < g(x)$ for every $x \in A_\alpha$, and $h(x) < f_\alpha(x)$ for every $x \in \overline{B}_\alpha$. Put $f(x) = \sup_\alpha f_\alpha(x)$; this function is indeed continuous on E . For every $x \in E$, there exists a neighborhood V of x meeting only finitely many of the A_α ; on this neighborhood f is therefore the upper envelope of *finitely* many distinct continuous functions, and hence is continuous. On the other hand, there exists an index α such that $x \in \overline{B}_\alpha$, so $f(x) \geq f_\alpha(x) > h(x)$, and there exists an index β such that $f(x) = f_\beta(x)$. Since $f(x) > h(x) \geq -\infty$, we necessarily have $x \in A_\beta$, and hence $f_\beta(x) < g(x)$, completing the proof.

Using this result, one can show that all signs $<$ in Theorem 9 may be replaced by $\leq$ without affecting the validity of the statement. Indeed, we may restrict ourselves to the case where $-1 \leq h(x) \leq g(x) \leq 1$ for every $x \in E$; otherwise, we would replace g by $\frac{g}{1+|g|}$ and h by $\frac{h}{1+|h|}$. We then define inductively three sequences (f_n) , (g_n) , (h_n) of numerical functions on E by the following conditions:

$$1^\circ \quad g_0(x) = g(x) + 1, \quad h_0(x) = h(x) - 1;$$

$$2^\circ \quad f_n \text{ is continuous on } E \text{ and satisfies } h_n(x) < f_n(x) < g_n(x);$$

$$3^\circ \quad g_n(x) = \text{Min} \left[g(x) + \frac{1}{2^n}, f_{n-1}(x) + \frac{1}{2^n} \right], \quad h_n(x) = \text{Max} \left[h(x) - \frac{1}{2^n}, f_{n-1}(x) - \frac{1}{2^n} \right].$$

An immediate induction shows that g_n is lower semicontinuous on E , h_n is upper semicontinuous, and $h_n(x) < g_n(x)$ for every x ; by Theorem 9, this shows that the definition can be continued for every n . Moreover, it is clear that $|f_n(x) - f_{n-1}(x)| \leq \frac{1}{2^n}$; hence the sequence (f_n) converges uniformly on E to a function f which is therefore continuous and clearly satisfies $h \leq f \leq g$.

12. To conclude, we indicate a number of problems raised by the questions treated in this paper:

I. We have seen (Theorem 4) that every metrizable space with a countable base is paracompact: *does this property extend to all metrizable spaces?* One could doubtless answer this question if a direct proof of Theorem 4 were found, but such a proof does not seem easy.

II. By Theorem 2, a product $E \times F$ of two topological spaces can be paracompact only if each of the spaces E , F is paracompact (these spaces being homeomorphic to closed subspaces of $E \times F$). Conversely, *is the product of two paracompact spaces necessarily paracompact?*

III. *Does Theorem 8 extend to all normal spaces?* In any case, it remains valid for the normal nonparacompact space E considered in Section 3⁽⁵⁾.

IV. *For a paracompact space, is the universal structure always a complete uniform structure?* This is so for locally compact σ -compact spaces and for metrizable spaces; on the other hand, the universal structure of the nonparacompact space E of Section 3 is not a complete uniform structure⁽⁵⁾.